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Operational performance and cost–performance trade-offs of low-cost, farmer-scale biochar kilns

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ABSTRACT

Smallholder adoption of biochar remains constrained by the limited availability of affordable, reliable, field-validated production technologies. Here we evaluated the operational performance, repeatability, and cost–performance trade-offs of three locally fabricable, low-cost, farmer-scale biochar kilns spanning a gradient of oxygen control: an open flame-curtain kiln (Type A; Traditional Oregon Kiln), a controlled-draft flame-curtain kiln (Type B), and an indirect-heat retort (Type C). Sixty batch trials ($n = 5$ per kiln × feedstock cell) used four representative Midwestern agroforestry residues. Mean gravimetric biochar yields differed sharply: 22.3% (Type A), 31.5% (Type B), and 46.5% (Type C), with corresponding mean peak temperatures of 510, 460, and 385 °C. A linear mixed-effects model (kiln fixed; feedstock and the kiln × feedstock interaction random), complemented by variance partitioning, identified kiln architecture as the dominant driver of yield, accounting for 92.4% of total variance ($\eta^2 = 0.924$; Cliff's $\delta = 1.00$ for all pairwise contrasts). Feedstock and the interaction together explained less than 1%, and the residual 6.7%. A generalized additive model showed that peak operating temperature alone explained 92% of the deviance in yield ($r = -0.957$, $p < 10^{-32}$); processing time was a much weaker, design-driven correlate. Yield repeatability was highest in Type B (CV = 6.8%), comparable in Type C (CV = 7.6%), and considerably lower in Type A (CV = 13.2%). Total fabrication and tooling costs ranged from about US$360 (Type A) to US$1280 (Type C). These benchmarks support evidence-based selection across a "technology ladder" of low-cost kilns and underpin wider deployment of decentralized, farmer-scale biochar systems.

1. Introduction

Biochar, a carbon-rich solid produced by pyrolyzing biomass under conditions of restricted oxygen availability, is increasingly recognized as a low-cost, multi-functional material with applications well beyond soil improvement [1–3]. This oxygen restriction is achieved either through open-air flame-curtain combustion, which consumes oxygen at the combustion front, or through sealed-chamber exclusion [4,5]. Beyond enhancing soil structure, nutrient retention, and long-term carbon storage, biochar offers practical pathways for residue management, decentralized carbon retention, and low-cost thermal energy use in resource-constrained settings. Global assessments estimate that sustainable biochar deployment could sequester up to 3 billion metric tons of CO_2 annually, highlighting its relevance for both climate mitigation and rural livelihoods [6]. Also, economic analyses suggest that small-scale biochar systems can be viable when capital costs are low and yields exceed roughly 30% [7]. Biochar can also reduce fertilizer demand by 10%–20%, saving USD $150–200 per hectare per year [8,9], while interest in carbon credits and ecosystem service payments continues to grow [10,11].

Across major agricultural regions such as the U.S. Midwest, 230–270 million metric tons of crop residues and wood by-products are generated annually, much of which remains underutilized or is openly burned [12,13]. Corn stover, sorghum residues, and forestry offcuts represent abundant, seasonally reliable feedstocks that could be converted into biochar to support residue valorization, reduce waste, and provide soil amendments or carbon-rich products for emerging markets. Agronomic studies indicate that biochar can reduce nitrous oxide

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emissions by 30%–50% [14,15] and improve soil water retention by up to 15% in typical Midwestern soils [9,16,17], although these effects represent only a fraction of biochar's broader utility.

Despite these benefits, adoption among smallholder and decentralized producers remains low. Commercial pyrolysis systems typically cost USD $5000–200,000 or more, far beyond the reach of smallholders, community groups, and small enterprises [18,19]. As a result, many farmers continue to rely on open burning or traditional earth pits, which yield only 10%–15% biochar, generate substantial smoke, and lose most feedstock carbon [20,21]. Limited training in controlled pyrolysis, temperature management [22], and quenching practices further constrains product quality and reproducibility [23]. Extension-oriented practice guidelines have begun to address this gap by documenting affordable, farmer-operable production methods and indicative production economics for small-scale and on-farm systems [24].

A central determinant of farmer-scale biochar production efficiency is the degree of oxygen control embedded in the kiln architecture. Oxygen availability governs the stability of the combustion–pyrolysis interface, which in turn controls peak operating temperature, volatile oxidation, and carbon retention [4,7]. Open flame-curtain kilns are widely accessible but exhibit variable airflow and fluctuating heat fronts, often resulting in incomplete volatile combustion and lower, more variable yields [19]. In these systems a sustained flame across the top of the feedstock pile consumes oxygen that would otherwise reach the char beneath, while the closed bottom and sides preserve the accumulating char until quenching [25]. Controlled-draft systems represent the next level of regulation, stabilizing the flame curtain and limiting entrainment of unburned particulates [7,26]. Indirect-heat retorts, exemplified by the Dartmoor Dragon Retort (DDR), provide still greater thermal consistency by isolating feedstock from direct combustion; although typically stationary, they enable oxygen-limited pyrolysis and achieve higher carbon conversion efficiencies [5,27]. These designs form a continuum of increasing oxygen control, from uncontrolled open burning, through flame-curtain and top-lit updraft systems, to indirect-heat retorts, along which yield, emissions, and process stability vary systematically. Table 1 summarizes representative small-scale biochar production technologies along this gradient, and the three archetypes evaluated in this study map directly onto its open flame-curtain, controlled-draft flame-curtain, and indirect-heat retort regions.

Few studies, however, have compared these architectures under field conditions using standardized feedstocks, leaving limited empirical guidance for decentralized, smallholder applications. Most prior work has characterized a single kiln design, evaluated reactors at laboratory or pilot scale, relied on non-standardized or residue-specific feedstocks, or framed the progression from open burning to retorts as a qualitative "technology ladder" without quantifying how performance changes from one rung to the next [7,19,26,32,35]. The absence of validated operational benchmarks for farmer-scale kilns limits the ability of extension programs, practitioners, and small producers to make informed decisions about technology selection, fabrication investment, and expected performance [7,24]. The present study addresses this gap by examining three affordable, farmer-scale biochar kiln archetypes that can be fabricated with basic tools and operated with minimal training. To our knowledge, it is among the few farmer-scale field studies to conduct a replicated, side-by-side comparison of open flame-curtain, controlled-draft flame-curtain, and indirect-heat retort kilns using standardized feedstocks under identical field conditions, rather than comparing results across studies that differ in reactor, scale, and feedstock. Building on this design, it is also among the first farmer-scale studies to partition yield variance between kiln architecture and feedstock, to apply generalized additive models (GAMs) that isolate peak operating temperature as the dominant thermal driver of yield, and to convert the qualitative technology ladder into a quantitative one that explicitly links fabrication cost, oxygen control, yield, and repeatability. By including both agricultural residues and agroforestry wood, the experiment further offers broader generalizability across common Midwestern feedstock types than prior residue-specific kiln evaluations, providing field-based performance benchmarks relevant to decentralized residue management and smallholder adoption.

1.1. Aim and objectives

This study delivers experimentally validated performance benchmarks for three farmer-scale biochar kiln archetypes that differ in oxygen control and operational stability. Performance was evaluated through three lenses: (i) production efficiency (gravimetric yield), (ii) operational throughput (batch cycle time), and (iii) process stability (repeatability). The specific objectives were to:

1. Characterize the technical operation of three kiln architectures using four common Midwestern agroforestry residues;
2. Quantify the trade-offs among gravimetric yield, processing time, and fabrication cost; and
3. Identify the drivers of process stability and yield repeatability to inform technology selection in resource-constrained settings.

Taken together, the replicated, multi-architecture field comparison and the accompanying statistical treatment are designed to address the gaps identified above directly: by partitioning variance across a fully crossed design, isolating the thermal driver of yield, and pricing each rung of the technology ladder, the study converts a largely qualitative literature into quantitative, design-anchored evidence on variance components, thermal drivers, and the cost–performance trade-offs that govern technology selection. To our knowledge, this combination of evidence has not previously been assembled at farmer scale.

2. Materials and methods

2.1. Study site

The study was conducted in West Lafayette, Tippecanoe County, Indiana, USA (approximately 40.4° N, 86.9° W). The region lies within the eastern U.S. Corn Belt and is characterized by gently undulating glacial till plains, fertile Mollisols and Alfisols that support intensive row-crop agriculture and reliable biomass availability [36,37]. The climate is humid continental (Köppen Dfa), with mean annual precipitation of approximately 1000 mm and monthly temperatures ranging from approximately 10 °C in January to 30 °C in July [38].

Kiln fabrication was carried out at a university research and machining facility, kiln testing and biochar production at a university forestry research farm, crop residues were collected at a university agricultural research center, and woody biomass was sourced from an associated university research forest. All sites are within approximately 10 km of one another, minimizing logistical variability and ensuring that feedstock handling, moisture exposure, and transport times reflected realistic smallholder conditions in the U.S. Midwest. Full descriptions of the four feedstocks used, including species, form, billet dimensions, and drying regime, are provided in Section 2.4 (Biochar production).

2.2. Kiln archetypes and selection rationale

Three legacy kiln concepts were selected to span the dominant modes of oxygen control found in smallholder-accessible systems:

- **Type A (TOK; open flame-curtain).** A widely used Traditional Oregon Kiln selected as the baseline for low cost, portability, and ease of operation. Limitations include uncontrolled airflow, fluctuating heat fronts, and variable yields [7,19]. Its rectangular fire-pan geometry, fabrication approach, and bill of materials follow widely circulated open-source shop drawings for the forestry-size Oregon Kiln [39].

Table 1
Characteristics of representative small-scale biochar production technologies, spanning the gradient of oxygen control from open burning to indirect-heat retorts.

Attribute	Open burn	Kon-Tiki, Oregon and Ring of Fire kilns	Top-lit updraft units (TLUD)	Metal kilns	Dartmoor Dragon retort (DDR)
Region availability	Worldwide, especially smallholder agriculture in Africa and Asia	United States, Asia and Africa	Latin America and Asia	Latin America, Europe, Asia and Africa	Latin America, Asia and Africa
Raw material	Small diameter wood, agro-industrial residues and crop stubble	Wood residues, bamboo, crop stubble and forest slash	Wood chips or similar small fractions	Wood, bamboo and medium sized agro-industrial residues	Wood, bamboo and fine agro-industrial residues such as coffee husk, sawdust, cotton stalks and straw
Typical yield (%)	10–15	Not formally established; generally higher than open burn	Not formally established; described as highly efficient	About 25–40	About 30–40
Typical batch time	Minutes to hours, dependent on feedstock size	Minutes to hours, depending on feedstock and kiln size	Minutes to hours, depending on batch size	About 2–8 h	Two hours or more
Emissions and air quality	Not controlled; high particulate and VOC release	Flame curtain reduces visible smoke; emissions partly uncontrolled	Controlled; relatively clean combustion	Controlled; reduced smoke and pollutants	Controlled; cleaner exhaust than open systems
Investment cost (order of magnitude)	Very low; little or no equipment cost	Inexpensive; fabricated from mild steel or dug as cones	Relatively inexpensive; small metal or ceramic units	High capital cost; from about 5000 US dollars to several million for large plants	Relatively expensive; about 500–800 US dollars for a 900 kg unit, more for larger plants
Scale and applicability	Small-scale, informal production near fields	Small-scale field production and demonstration	Small-scale household or farm-scale production	Small to large-scale, including industrial plants	Small to large-scale depending on retort size
Mobility	No; cannot be moved	Yes; portable metal cones or excavated pits	Yes; portable stove-like units	Yes or no, depending on design	Yes or no; skid-mounted retorts may be moved
Labor intensity	High; continuous manual feeding	Lower than open burns; still requires active feeding and quenching	Low to moderate	Low; batch loading and unloading	Low; controlled firing and batch unloading
Air and gas circulation control	Not controlled; fully open	Partially controlled by flame curtain; no sealed chamber	Controlled primary and secondary air	Controlled via ports and chimneys	Controlled through sealed retort and flue system
Durability	Low; no permanent structure	Depends on material; steel cones last several seasons	Very durable; simple maintenance	Very durable; repairable steel	Very durable; modular steel parts replaceable
Typical lifespan	Single events or short term	A few months to several years, depending on material	Long-term with basic maintenance	At least two to three years, often longer	Long-term; components replaceable
References	Cornelissen et al. [7], Dam [21]	Schmidt and Taylor [28], IBI [29], Asare [30]	Shankar [31], Brown [32]	Mohan et al. [33]	Biochar-Retort [34], Antal et al. [4], Brewer et al. [5]

- **Type B (CDFC; controlled-draft flame-curtain).** A design developed in this study that builds on the established Ring-of-Fire flame-curtain concept, upgrading it with a fitted lid, controlled air inlets, and a chimney vent to stabilize draft and the combustion–pyrolysis interface [7,26]. It consists of a steel base and a fitted lid, designed and fabricated in this study on the Type A footprint.
- **Type C (IHFRt; indirect-heat fine-feedstock retort).** An adaptation of the Dartmoor Dragon Retort, fabricated as a fully-steel mobile unit with an integrated sealed inner charge chamber to enable oxygen-limited pyrolysis [5,27].

These three farmer-scale biochar production systems are shown in the Supplementary Information (Figure S1).

Table 2 compares them across the design attributes that motivated their inclusion: degree of oxygen control, characteristic thermal behavior, mobility, and fabrication complexity. The comparison uses conceptual labels and design-level distinctions only; measured operational outcomes are reported in the Results.

The three concepts were intentionally designed as an additive system in which each successive kiln incorporates the structure of the previous one. This reflects realistic fabrication pathways in smallholder or extension settings, where users may progressively upgrade an existing kiln rather than construct entirely new units. A detailed list of materials, tooling, and equipment is provided in the Appendix (Table 7); open-source shop drawings and indicative fabrication costs for the Type A Oregon Kiln baseline are available in published practice guidelines [39], whereas the Type B and Type C designs were developed in this study.

2.3. Fabrication and dimensions

The dimensions of Type A follow the bill of materials and shop drawings published for the forestry-size Oregon Kiln [39], which served as the open-source baseline for this study. Types B and C are original designs developed here: each was sized additively on the preceding unit, with Type B built on the Type A footprint and Type C built on the Type B configuration, so that the Type A dimensions establish the common footprint from which the two higher-tier designs were derived. All three kilns were fabricated from locally available mild steel using a standardized workflow. Type A is a single open rectangular vessel

Table 2
Design-level comparison of the three kiln archetypes selected for this study. Attributes reflect the selection rationale (oxygen control, thermal behavior, and fabrication considerations) rather than measured performance, which is reported in the Results.

Attribute	Open flame-curtain (Type A)	Controlled-draft flame-curtain (Type B)	Indirect-heat retort (Type C)
Oxygen control	Open top; uncontrolled air entry	Fitted lid and controlled inlets; chimney vent	Sealed inner chamber; minimal air entry
Thermal behavior	Highly variable; fluctuating heat fronts	Moderately stable; improved draft	Most stable; indirect heating
Typical temperature	>500 °C [19]	Mid-400 °C [7,40]	<400 °C [5]
Strengths	Portable; fast batches; easy to operate	Reduced smoke; more stable flame curtain	Consistent performance; oxygen-limited pyrolysis
Weaknesses	Variable yields; smoke; incomplete combustion	More fabrication effort; approximately twice the weight of Type A	Longest batch time; highest material requirement
Mobility	Fully portable	Portable with moderate weight	Mobile in this study; traditional retorts are stationary
Fabrication complexity	Very low	Moderate	Highest

(1.12 m × 1.02 m × 0.33 m; 44 in × 40 in × 13 in; usable volume ≈ 0.376 m^3, or ≈ 376 L). Type B retains the Type A footprint with an additional upper extension and fitted lid that doubles the wall height to 0.66 m (26 in; usable volume ≈ 0.754 m^3, or ≈ 754 L). Type C retains the Type B configuration and integrates a sealed inner chamber (0.36 m^3; 14 in; chamber volume ≈ 0.0467 m^3, or ≈ 46.7 L) supported within the outer firebox, creating an annular combustion volume of ≈ 0.707 m^3, or ≈ 707 L. The materials and tooling used for fabrication are listed in the Appendix (Table 7) ; the Type A bill of materials follows the published open-source Oregon Kiln drawings [39], while the additional materials for Types B and C reflect the design modifications introduced in this study.

2.4. Biochar production

Biochar production trials were conducted at a university forestry research farm. Four feedstocks representative of Midwestern agroforestry and agricultural systems were used: corn cobs, corn stalks, sorghum residue, and red elm (*Ulmus rubra*) wood. Crop residues were collected post-harvest and air-dried for approximately 7 days; dead elm stems were bucked into 10 in billets, split, and air-dried for approximately 14 days, all under covered but ventilated conditions. Red elm was harvested as standing deadwood from an associated university research forest and had already senesced and stabilized prior to collection. Feedstock preparation followed the size and moisture criteria recommended for flame-cap operation, with pieces kept within roughly the 1–4 in range and target moisture below about 25% (wet basis) to sustain an even flame front and limit smoke [25].

Each kiln–feedstock combination was run five times, yielding 60 independent batches in a fully crossed two-factor factorial design. For each run, initial feedstock mass was measured using a calibrated 125 kg floor scale. Operating procedures differed by kiln type to reflect their distinct airflow configurations. For the open and controlled-draft flame-curtain kilns (Types A and B), an initial rick was loaded and ignited evenly across the top, and subsequent layers of similarly sized material were added as each preceding layer developed a film of white ash, maintaining a flame cap that suppressed smoke until the vessel filled with hot char; burns were then terminated by water quenching or, for Type B, by sealing the fitted lid to exclude air [25].

For the indirect-heat retort (Type C), the inner sealed charge chamber was filled with prepared feedstock and closed, and a separate charge of firewood was loaded into the surrounding annular space of the outer firebox to provide the external heat source. The firewood bed was ignited from above and managed to maintain a stable combustion front around the inner chamber, with no direct flame or flue connection to the biochar charge.

Kiln operating temperatures were monitored with a four-channel K-type thermocouple data logger (Landtek TM-902C or equivalent), with stainless-steel sheathed thermocouples (3 mm diameter, 1 m length) inserted through access ports at upper, mid-depth, lower, and near-wall positions. The data logger's K-type measurement range is −200 to 1372 °C, with a manufacturer-specified accuracy of ±0.2% of reading plus 0.7 °C and a display resolution of 0.1 °C. Temperatures were logged at 10 s intervals from ignition through quenching. Feedstock moisture content was determined gravimetrically at 105 °C to constant mass, following ASTM E1756 (Standard Test Method for Determination of Total Solids in Biomass). Gravimetric biochar yield was computed as

$$\text{Yield (\%)} = \frac{\text{Mass of biochar}}{\text{Mass of biomass}} \times 100. \quad (1)$$

2.5. Statistical analysis

The analysis followed a two-step strategy. First, the assumptions and structure of the data were diagnosed; second, the structure was matched with appropriate parametric and semi-parametric models. All analyses were conducted in R 4.4 (`lme4`, `lmerTest`, `performance`, `emmeans`, `mgcv`, `boot`, `rstatix`, `effectsize`; `pingouin` and `statsmodels` were used in Python for cross-validation). The analysis code and cleaned dataset are available from the authors on reasonable request.

Diagnostics: Within-cell normality was assessed using Shapiro–Wilk tests [41] for each of the twelve kiln × feedstock cells (n = 5 each); homogeneity of variance across kilns was assessed with Levene's test based on medians [42].

Variance partitioning and effect sizes: Differences in biochar yield were modeled with a two-way fixed-effects analysis of variance (ANOVA), with kiln, feedstock, and their interaction as fixed effects. Effect sizes were summarized as eta-squared (η^2) and partial omega-squared (ω_p^2) [43]. Welch's heteroscedasticity-robust ANOVA [44] was computed to verify robustness of the main kiln effect under unequal variances. To complement the fixed-effects decomposition, a crossed random-effects model

$$Y_{ijk} = \mu + b_i^{\text{kiln}} + b_j^{\text{feed}} + \varepsilon_{ijk}, \quad b_i^{\text{kiln}} \sim \mathcal{N}(0, \sigma^2_{\text{kiln}}), \quad b_j^{\text{feed}} \sim \mathcal{N}(0, \sigma^2_{\text{feed}}),$$
$$\varepsilon_{ijk} \sim \mathcal{N}(0, \sigma^2_{\text{res}})$$

was fit by restricted maximum likelihood (REML) to estimate the variance components attributable to kiln and feedstock under a generalizable framework. A linear mixed-effects model with kiln as a fixed effect and feedstock as a random effect was additionally fit to obtain population-level kiln estimates corrected for feedstock-level grouping.

Pairwise comparisons: Pairwise differences among kilns were evaluated using Tukey's HSD [45] (parametric, family-wise error rate α =

Table 3
Descriptive statistics for the 60 batch trials (mean (SD); medians for processing time given the bimodal feedstock split). CI denotes the bootstrap 95% confidence interval of the mean (10,000 resamples).

Metric	Type A (TOK)	Type B (CDFC)	Type C (IHFRt)
n (batches)	20	20	20
Yield (%, dry basis)	22.32 (2.94)	31.52 (2.16)	46.51 (3.53)
Yield 95% bootstrap CI	21.08–23.61	30.59–32.43	44.88–47.90
Peak temperature (°C)	510.3 (5.6)	460.0 (5.5)	385.3 (9.6)
Processing time, min (median)	41.1 (10)	53.8 (32)	127.9 (50)
Moisture content (%)	14.6 (7.0)	13.6 (6.6)	13.0 (4.7)
Coefficient of variation, yield (%)	13.16	6.85	7.60

0.05) and, in parallel, a non-parametric Dunn-style test [46] with Holm correction [47]. Cohen's d [48] and Cliff's δ [49] were computed for each pairwise contrast as parametric and rank-based effect size measures, respectively.

Repeatability: Process stability was characterized by (i) within-cell coefficient of variation (CV) averaged across feedstocks, (ii) overall within-kiln CV, and (iii) the intraclass correlation coefficient (ICC) [50] representing the share of within-kiln variance attributable to feedstock-level grouping versus residual batch-to-batch noise.

Bootstrapped intervals: Non-parametric percentile confidence intervals (95%) for kiln means and pairwise mean differences were computed from 10,000 bootstrap resamples [51] (seeded for reproducibility).

Functional drivers: The relationship between gravimetric yield and continuous predictors (peak operating temperature, processing time, feedstock moisture content) was examined with both linear models and generalized additive models (GAMs) using thin-plate regression splines (`mgcv::s()`, default smoothing parameter selection by restricted maximum likelihood (REML)) [52]. GAMs were fit globally and within each kiln to test whether functional relationships persisted after accounting for between-kiln differences. Models were compared by adjusted R^2 (proportion of variance explained, adjusted for the number of parameters; higher is better), generalized cross-validation (GCV) scores (approximate out-of-sample prediction error; lower is better), and Akaike information criterion (AIC; likelihood-based model selection score; lower is better).

3. Results

3.1. Kiln-level operational performance

Across the 60 batch trials, kiln architecture produced sharply divergent operational performance (Fig. 1; Table 3). Mean gravimetric biochar yield rose monotonically from Type A (mean 22.32%, 95% bootstrap CI 21.08–23.61) to Type B (31.52%, CI 30.59–32.43) to Type C (46.51%, CI 44.88–47.90). Mean peak operating temperature decreased over the same gradient, from 510.3 °C (SD 5.6) in Type A to 460.0 °C (SD 5.5) in Type B and 385.3 °C (SD 9.6) in Type C, while mean batch processing time increased from 41.1 min in Type A (median 10) to 53.8 min in Type B (median 32) to 127.9 min in Type C (median 50). The latter two distributions are bimodal because crop residues required short cycles (~10–60 min) whereas solid wood required substantially longer cycles (~2–6 h) within each kiln.

3.2. Variance partitioning: kiln architecture is the dominant driver

A two-way fixed-effects ANOVA confirmed a highly significant kiln effect on yield ($F_{2,48} = 330.3$, $p < 10^{-25}$), with non-significant feedstock and interaction effects ($p > 0.4$ in both cases). Variance partitioning attributed 92.4% of the total sum of squares to kiln, 0.7% to feedstock, 0.2% to the kiln × feedstock interaction, and 6.7% to residual batch-to-batch noise (Fig. 2a). The crossed random-effects model produced a consistent picture: estimated variance components were $\hat{\sigma}^2_{\text{kiln}} = 148.6$ (94.5%), $\hat{\sigma}^2_{\text{feed}} = 0.5$ (0.3%), and $\hat{\sigma}^2_{\text{res}} = 8.2$ (5.2%) (Fig. 2b). Effect-size estimates ($\eta^2_{\text{kiln}} = 0.924$; partial $\omega^2_{\text{kiln}} = 0.917$) confirmed that kiln architecture predominantly dominated yield variation. Welch's heteroscedasticity-robust ANOVA produced consistent inferences ($F_{2,36.4} = 272.6$, $p < 10^{-21}$).

Table 4
Pairwise comparisons among kilns: bootstrapped mean differences ($\Delta\bar{Y}$) with 95% confidence intervals, Tukey HSD adjusted p-values, Cohen's d, and Cliff's δ. All contrasts are statistically significant after multiple-comparison adjustment, with extreme effect sizes.

Contrast	$\Delta\bar{Y}$ (%) [95% CI]	Tukey p-adj	Cohen's d	Cliff's δ
A − B	−9.20 [−10.71, −7.63]	$< 10^{-6}$	−3.57	−0.99
A − C	−24.20 [−26.12, −22.19]	$< 10^{-9}$	−7.44	−1.00
B − C	−14.99 [−16.68, −13.13]	$< 10^{-7}$	−5.12	−1.00

3.3. Pairwise contrasts across the technology ladder

All three pairwise contrasts among kilns were both statistically and practically significant (Table 4). Mean yield differences were 9.2% (Type A vs. B), 24.2% (A vs. C), and 15.0% (B vs. C), all with non-overlapping bootstrap 95% confidence intervals. Cohen's d values (−3.6 to −7.4) were extreme, and the rank-based Cliff's δ reached the maximum value of 1.00 in every kiln contrast: every batch from a higher-tier kiln out-yielded every batch from a lower-tier kiln. Tukey HSD and a Dunn-style non-parametric test produced concordant inferences ($p < 10^{-7}$ in all contrasts; Table 4).

3.4. Feedstock effects and operational factors

Although kiln architecture dominated, feedstock identity exerted measurable secondary effects on processing time and within-cell repeatability rather than on mean yield (Fig. 3). Within each kiln, mean yields varied across feedstocks by less than 4 percentage points, and crop residues, despite differing in particle size and bulk density, produced comparable yields (Fig. 3). Solid wood at the highest moisture contents (19%–27%) yielded slightly lower than crop residues in the open and controlled-draft kilns but matched crop-residue yields in the indirect-heat retort, indicating that thermal isolation effectively decouples performance from feedstock moisture (within-kiln Pearson correlation of moisture and yield: Type A $r = -0.32$, Type B $r = -0.47$, Type C $r = -0.22$; only Type B reached $p < 0.05$).

3.5. Functional drivers: temperature dominates, time is design-driven

Across all batches, peak operating temperature was the single strongest predictor of yield (Pearson $r = -0.957$, $p = 5.6\times10^{-33}$; Fig. 4a). A generalized additive model with a thin-plate spline of temperature explained 93.1% of the deviance in yield (GCV = 11.0), with a near-monotonic decline of approximately 0.19 percentage points of yield per °C across the experimentally observed range (~370–525 °C). The GAM that additionally included a smooth term for processing time produced only a marginal improvement (deviance explained = 0.935), confirming that the temperature effect captures most of the operationally relevant signal.

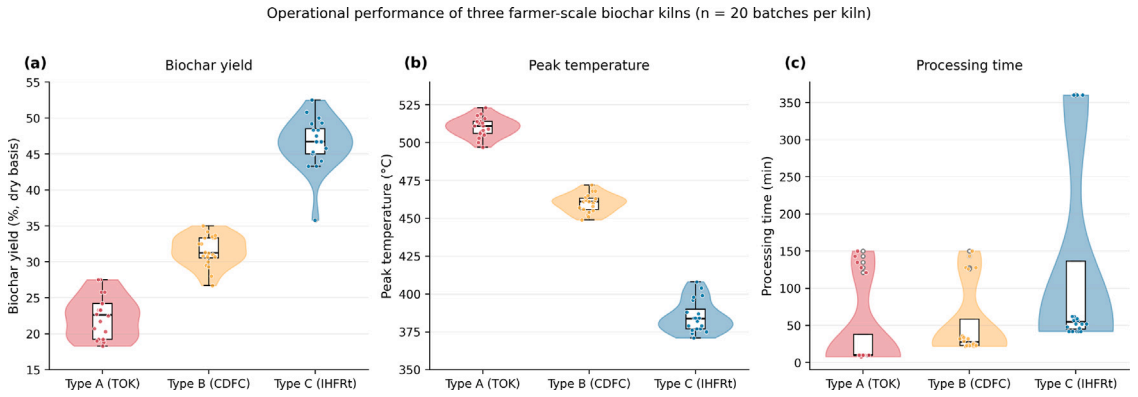


Fig. 1. Operational performance of three farmer-scale biochar kilns. Violins show the distribution of (a) gravimetric biochar yield, (b) peak operating temperature, and (c) processing time across 20 batches per kiln (5 replicates × 4 feedstocks). Inner boxplots indicate the median and interquartile range; jittered points show individual batches.

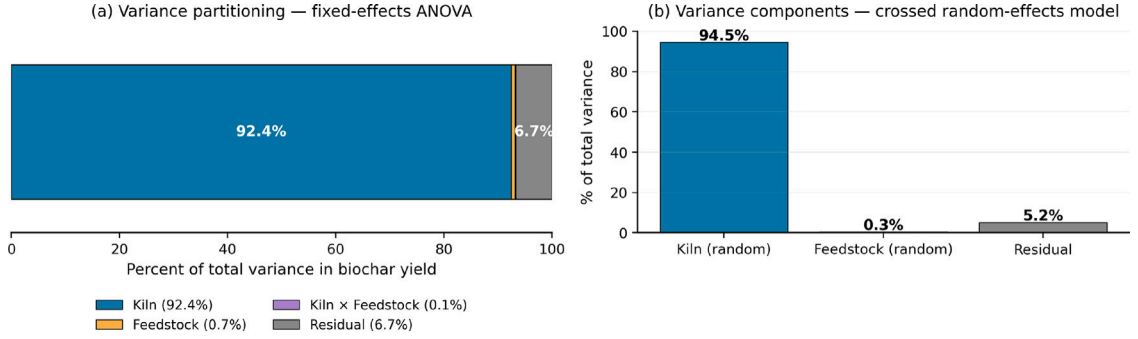


Fig. 2. Variance partitioning of biochar yield. (a) Stacked decomposition of the total sum of squares from a two-way fixed-effects ANOVA. (b) Estimated variance components ($\hat{\sigma}^2$) from a crossed random-effects model. Both methods identify kiln architecture as the overwhelmingly dominant source of variation in yield.

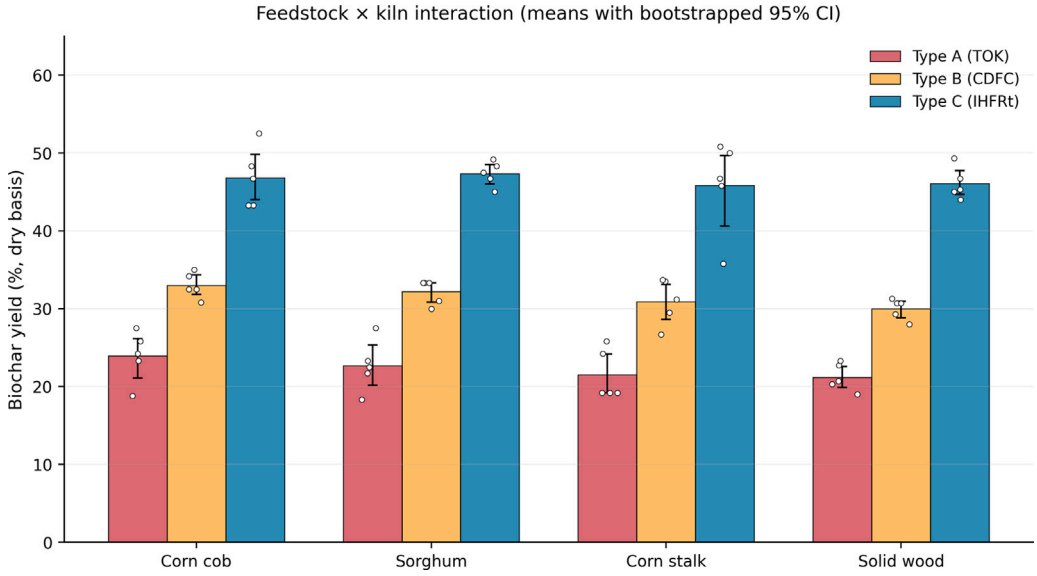


Fig. 3. Feedstock × kiln interaction in biochar yield. Bars show cell means with bootstrapped 95% confidence intervals; jittered points show individual batches. The kiln effect is consistent across feedstocks; the feedstock effect within kilns is small and largely confined to solid wood at higher moisture contents.

In contrast, the across-batch correlation between yield and processing time was modestly positive ($r = +0.332$, $p = 0.010$; Fig. 4b). This sign reflects the kiln-segmented design: kilns that retain heat and restrict oxygen require longer batch times to complete pyrolysis but ultimately recover a greater mass fraction of the feedstock as solid product. Within kilns, processing time was a poor predictor of yield ($R^2 \leq 0.19$ in all three kilns), with no statistically significant slopes after correction. The trade-off between throughput and biochar recovery is therefore an emergent property of the architecture, not a within-kiln operational lever (Fig. 5).

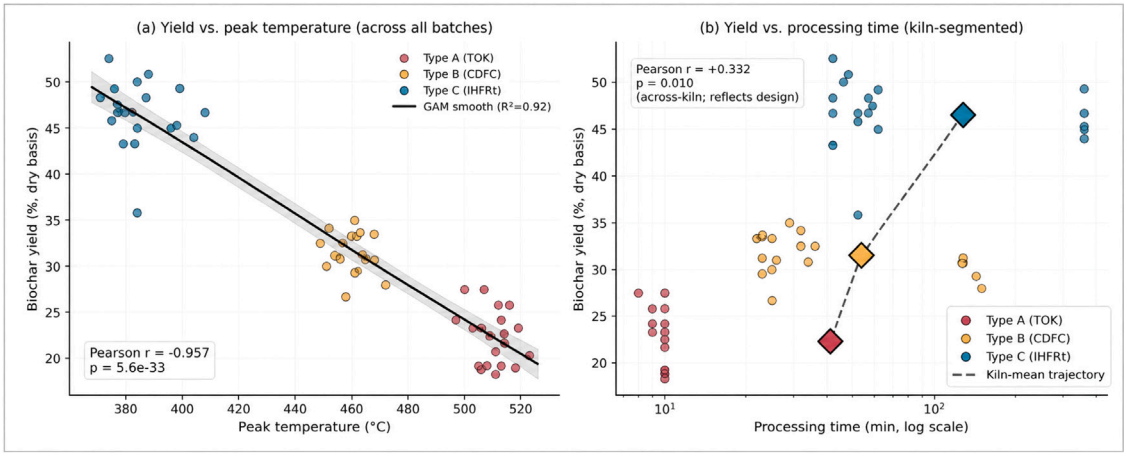


Fig. 4. Functional drivers of biochar yield. (a) Across all 60 batches, peak operating temperature explains 92% of the deviance in yield (GAM smooth, shaded 95% CI). (b) Yield versus processing time on a logarithmic scale; the dashed trajectory connects kiln-mean centroids. The aggregate positive correlation reflects the kiln-segmented design, not a within-kiln lever.

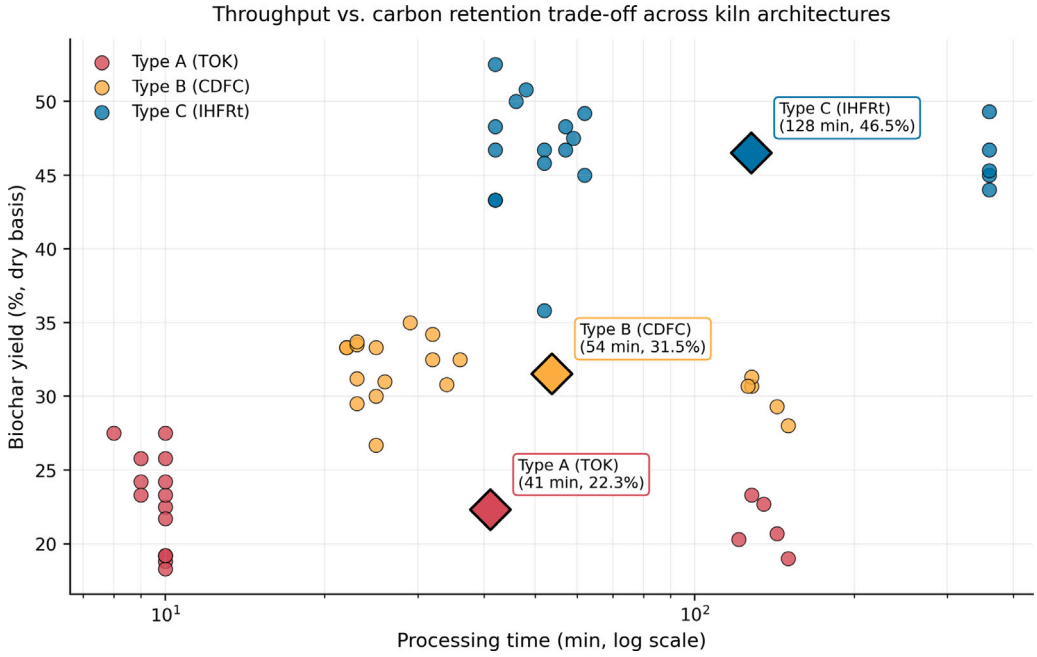


Fig. 5. Throughput × biochar-yield trade-off. Each circle is a single batch; diamonds mark kiln-mean centroids. Type A delivers the fastest throughput at the lowest yield, Type C the highest yield at the longest cycle time, and Type B occupies a Pareto-style middle position.

3.6. Repeatability and process stability

Batch-to-batch repeatability of yield differed substantially across the three kilns (Fig. 6; Table 5). The open flame-curtain kiln (Type A) was the least repeatable, with an overall coefficient of variation of 13.16% and a within-cell CV of 12.92%. The controlled-draft kiln (Type B) was the most repeatable, with an overall CV of 6.85% and a within-cell CV of 5.95%, indicating that even modest constraints on airflow translated into roughly half the batch-to-batch noise. The indirect-heat retort (Type C) was comparably repeatable (overall CV 7.60%; within-cell CV 7.28%) but slightly more dispersed than Type B, partly reflecting a single low-yielding corn-stalk batch. ICCs computed within each kiln (variance among feedstock means relative to total within-kiln variance) were 0.15, 0.31, and 0.03 for kilns A, B, and C, respectively. The very low ICC in Type C confirms that the retort effectively decouples performance from feedstock identity.

3.7. Cost and the technology ladder

Total fabrication and tooling costs (mild steel, welding consumables, ignition aids, personal protective equipment (PPE), quenching equipment, and quality-control instruments) ranged from approximately US$359 (Type A) to US$768 (Type B) and US$1282 (Type C), with structural-steel inputs accounting for roughly 30%–45% of total cost in each kiln. These figures are consistent with indicative fabrication costs reported for comparable open-source flame-curtain kilns of the Oregon Kiln class [39]. The three kilns therefore form a clear technology ladder spanning a factor-of-three range in capital cost, a factor-of-two range in yield, a factor-of-three range in batch cycle time, and a roughly twofold range in repeatability (Fig. 7; Fig. 8). A simple fabrication-cost comparison suggests that the additional cost of moving from Type A to Types B or C may be offset within a small number of additional production cycles, although this depends on local biochar prices, labor costs, and feedstock logistics.

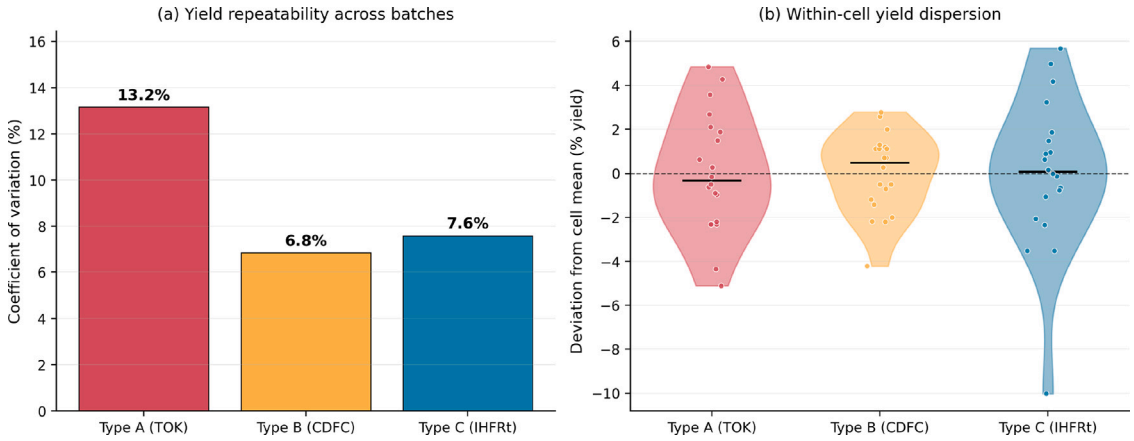


Fig. 6. Batch-to-batch repeatability of biochar yield. (a) Coefficient of variation across all batches per kiln. (b) Distribution of batch deviations from cell means (kiln × feedstock); tighter distributions indicate higher repeatability. The controlled-draft kiln is the most repeatable, the open flame-curtain kiln the least.

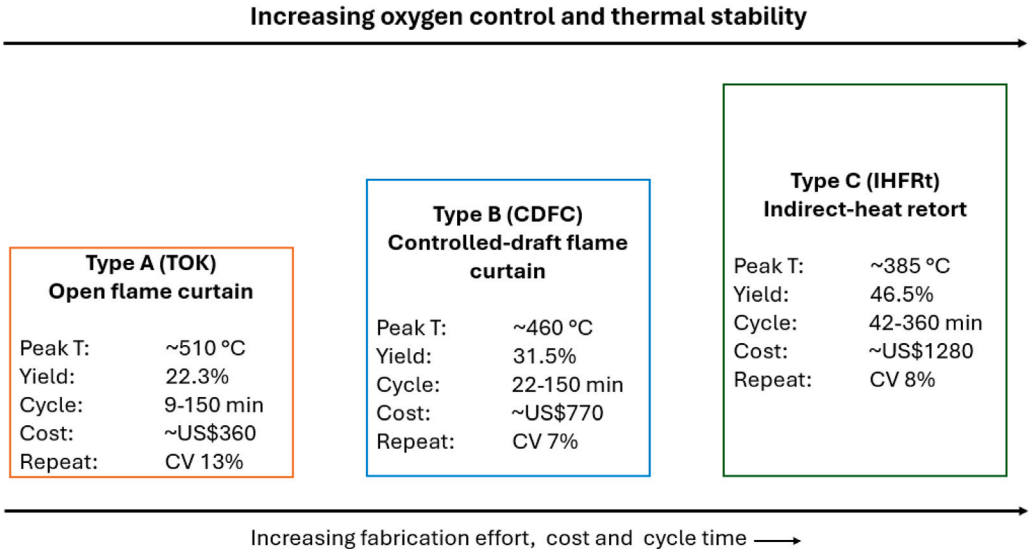


Fig. 7. The "technology ladder" for farmer-scale biochar production. Each successive design tightens oxygen control and thermal stability at the cost of increased fabrication effort and longer cycle times.

Table 5
Repeatability metrics by kiln. CV: coefficient of variation. ICC: intraclass correlation coefficient (variance among feedstock means as a share of total within-kiln variance).

Kiln	Yield mean (SD)	CV within-cell (%)	CV overall (%)	ICC (feedstock)
Type A (TOK)	22.32 (2.94)	12.92	13.16	0.147
Type B (CDFC)	31.52 (2.16)	5.95	6.85	0.313
Type C (IHFRt)	46.51 (3.53)	7.28	7.60	0.033

Overall, the patterns delineate a clear technology ladder in which capital cost and batch cycle time increase by roughly a factor of three, and biochar yield and repeatability increase by roughly a factor of two. To summarize the most decision-relevant attributes for smallholder use, Table 6 compares the three kilns across capital cost, yield, smoke, and feedstock compatibility.

4. Discussion

4.1. Kiln architecture and the oxygen-control gradient

The central result of this study is that kiln architecture, rather than feedstock identity, governs farmer-scale biochar production efficiency. Variance partitioning, fixed- and random-effects models, and a battery of effect sizes converged on a single conclusion: 92%–95% of the variation in gravimetric yield was attributable to kiln type, and every batch from a higher-tier kiln out-yielded every batch from a lower-tier kiln (Cliff's $\delta = 1.00$ for all contrasts). Several features of this contribution are, to our knowledge, new at farmer scale. Among the few farmer-scale field studies to place open flame-curtain, controlled-draft flame-curtain, and indirect-heat retort kilns side by side under standardized feedstocks and identical field conditions, the present work extends previous qualitative "technology ladder" framings by quantifying the magnitudes that earlier accounts left implicit: it partitions yield variance between architecture and feedstock, showing that kiln type accounts for over 90% of yield variation whereas feedstock contributes less than 1%, and it applies generalized additive models to isolate peak operating temperature as a single thermal driver explaining roughly 92% of the

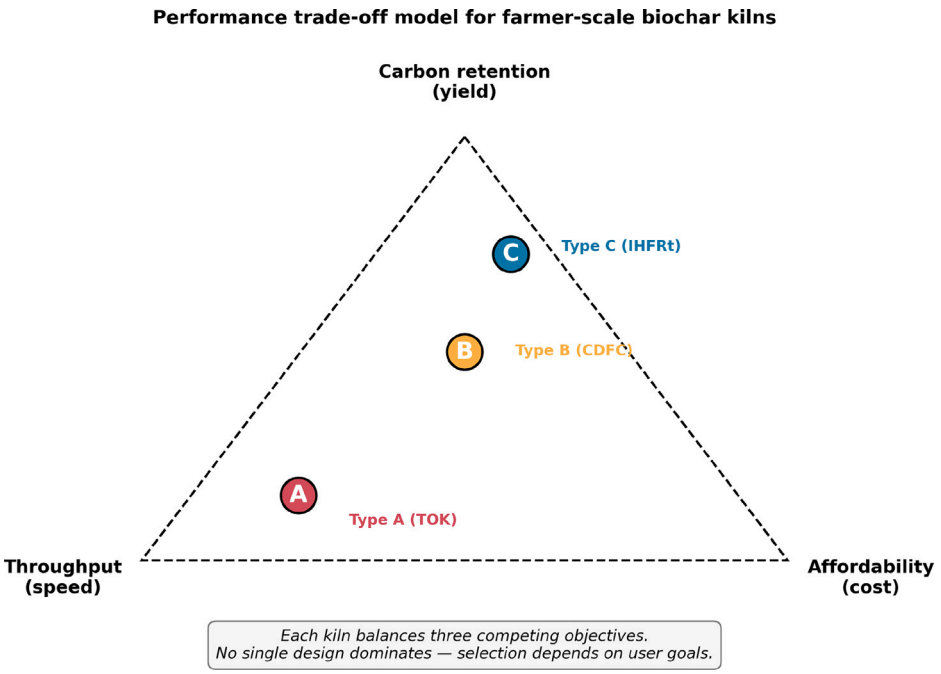


Fig. 8. Conceptual trade-off model. Each kiln balances three competing objectives: biochar yield, throughput, and affordability. No design dominates on all three axes; selection depends on user goals, feedstock supply, and capacity for fabrication and operation.

Table 6
Qualitative comparison of kiln performance characteristics relevant to smallholder use.

Kiln type	Capital cost (approx.)	Typical yield	Visible smoke/emissions	Feedstock compatibility
Type A	Lowest (US$360)	Lowest (baseline)	Highest; more variable flame and smoke plume	Compatible with all tested residues; more sensitive to moisture and packing
Type B	Moderate (US$760–770)	Intermediate (higher than Type A, lower than Type C)	Reduced smoke compared to Type A; more stable flame curtain	Compatible with all tested residues; moderate sensitivity to moisture
Type C	Highest (US$1280)	Highest	Lowest visible smoke; most controlled combustion zone	Compatible with all tested residues; least sensitive to moisture and particle size

deviance in yield. The importance of oxygen regulation is itself well established; [4] demonstrated that restricting oxidant access during flash carbonization sharply raises fixed-carbon yield, and [7] attributed the comparatively low yields of open flame-curtain kilns to volatile oxidation at the combustion front. What that literature has largely left unresolved is the relative weight of architecture against the many competing sources of variation, particularly feedstock, that operate simultaneously under field conditions. By partitioning variance across a fully crossed design, the present work supplies that missing quantity: across the three archetypes, design choice alone accounted for the dominant majority of yield variation, whereas feedstock and its interaction with kiln were statistically negligible. The functional analysis sharpens the interpretation further, in that peak operating temperature alone explained 92% of the deviance in yield, with a near-monotonic negative slope of approximately −0.19 percentage points per °C, indicating that the differences among kilns can be parsimoniously summarized by the thermal regime each architecture imposes.

This thermal regime, in turn, is set mechanistically by how each design admits and manages oxygen. Type A's open flame-curtain permits unrestricted oxygen ingress and sustained temperatures above 500 °C, conditions under which a substantial fraction of the carbon-rich pyrolysis intermediates is oxidized rather than retained in the solid product [26,53]. Introducing a fitted lid and chimney in Type B narrows the air-entry pathway and lowers the peak operating temperature by roughly 50 °C, shifting the combustion–pyrolysis interface toward more complete devolatilization; [26] reported that precisely this kind of draft regulation reduces both methane release and the oxidation losses that depress yield in unmanaged flame curtains. Type C extends the logic to its limit by isolating the charge entirely from the combustion zone, so that externally burning off-gases drive endothermic pyrolysis at the lowest observed peak operating temperature (~385 °C). That regime falls within the temperature window that [32] and [54] identify as optimal for solid-product formation in slow-pyrolysis systems, which explains why our retort yields (45.8–47.3%) approach those of small-scale industrial units despite its modest fabrication cost. These comparisons show that the present study does not merely reproduce the qualitative ordering anticipated by earlier work; it quantifies how strongly that ordering dominates competing factors and ties the ranking to a single, measurable thermal lever, which is directly actionable for anyone selecting or fabricating a farmer-scale kiln.

4.2. Reconciling the yield-time trade-off

A finding that warrants careful interpretation concerns the relationship between yield and processing time. Across all 60 batches the correlation is weakly *positive* ($r = +0.33$), the opposite of the negative association sometimes implied by summaries of the pyrolysis literature, where slower heating and longer residence times are linked to higher char yields within a single reactor [55]. The apparent contradiction

dissolves once the level at which each effect operates is made explicit. The positive aggregate sign is an artifact of the kiln-segmented design: Type C is simultaneously the highest-yielding and the slowest architecture, while Type A is both the fastest and the lowest-yielding, so pooling batches across kilns confounds an architectural ranking with a duration axis. Within any single kiln, where reactor design is held constant and the residence-time argument of Bridgwater [55] should apply, processing time is in fact a poor predictor of yield ($R^2 \leq 0.19$), and no within-kiln slope remains statistically significant after correction.

The mechanistic reason is that duration and architecture are not interchangeable controls on the same variable. Longer cycle times in Type C are a consequence of indirect heat transfer and oxygen exclusion, not their cause; in this design, that indirect regime is sustained by a separate firewood charge in the outer combustion box, so the biochar feedstock in the sealed inner chamber is heated only by the surrounding combustion gases rather than by direct flame contact. The high yield arises because the charge never contacts the oxidizing flame front, whereas in Type A the same biomass is exposed to an oxygen-rich, high-temperature environment regardless of how the burn is paced. Extending a Type A batch therefore cannot reproduce Type C performance, because additional time in an oxidizing regime continues to consume the solid product rather than preserve it. The practical implication is that throughput and biochar recovery trade off against each other at the level of design choice rather than daily operation: an operator cannot recover the yield penalty of a low-control kiln by simply running it longer, and conversely, the slower cycle of a high-control kiln is the price of, not an obstacle to, its superior recovery. This reframing matters for extension guidance, which has at times advised prolonging burns to improve output, a strategy these data indicate is ineffective in the kiln class most accessible to smallholders.

4.3. Feedstock effects are second-order and moderated by kiln design

Feedstock identity contributed less than 1% of yield variance and never produced a within-kiln yield range exceeding four percentage points, a result that sits uneasily beside a substantial body of work reporting feedstock as a first-order determinant of biochar yield and quality. The meta-analysis of Jalali et al. [23], for instance, found that feedstock type and pyrolysis conditions jointly shaped the physical and chemical properties of biochars across a large compiled dataset and [35] likewise documented appreciable yield differences among feedstocks processed in farmer-relevant kilns. The discrepancy is real but not contradictory, and resolving it clarifies the scope of each claim. Studies that isolate strong feedstock effects typically hold the reactor and its temperature programme fixed, so that feedstock-driven differences in volatile content, lignin fraction, and bulk density are expressed directly in the product. In the present design, by contrast, the experimental contrast was the kiln itself, and the between-kiln differences in thermal regime were large enough (a peak-temperature span of roughly 125 °C) to swamp the comparatively modest feedstock signal. Feedstock effects are therefore not absent in our data; they are second-order relative to an architectural axis that earlier single-reactor studies did not vary.

The way feedstock influence changed across the technology ladder supports this reading mechanistically. The largest feedstock-driven differences appeared in Type A and Type B, where solid wood at elevated moisture (19%–27%) yielded slightly below the drier crop residues, consistent with the expectation that an exposed combustion front is sensitive to the energy diverted to water vaporization and to particle-size effects on packing and airflow. In Type C the effect vanished: yields were statistically indistinguishable across all four feedstocks, and the ICC for feedstock-level grouping fell to 0.03. Because the retort holds the pyrolysis zone at a controlled temperature behind a physical barrier, it buffers exactly the moisture- and morphology-driven variability that [55] is identified as central to feedstock sensitivity in directly heated systems, an interpretation also in line with the feedstock-and-kiln comparisons of Sarpong et al. [35]. The practical consequence is a form of feedstock flexibility that scales with the rung of the ladder: as oxygen control increases, the premium on tightly specified feedstock preparation falls, which lowers the operator-skill threshold for achieving acceptable, repeatable yields and broadens the range of residues a smallholder can process without sacrificing output.

4.4. Repeatability and the case for moderate complexity

The repeatability results invert an intuition that runs through much of the kiln-design literature, namely that tighter isolation of the charge should translate monotonically into more consistent output. Instead, the controlled-draft kiln (Type B) was the most repeatable design (overall CV 6.85%), narrowly outperforming the more thoroughly isolated indirect-heat retort (Type C, CV 7.60%), while both far surpassed the open flame-curtain kiln (Type A, CV 13.16%). The contrast between Type A and the two enclosed designs is mechanistically unsurprising and aligns with the rationale [7] advanced for draft regulation: constraining airflow stabilizes the flame curtain, narrows the range of local temperatures the biomass experiences, and so compresses batch-to-batch variation. The more informative comparison is between Type B and Type C, because it shows that beyond a threshold, additional architectural complexity buys mean yield rather than consistency.

Two mechanisms plausibly account for Type B's edge. First, repeatability is integrated over the duration of a batch, and Type C's substantially longer cycles expose each run to a wider window of fluctuation in operator intervention, ambient temperature, and wind, so that even a thermally well-buffered system accumulates more opportunities for run-to-run divergence; the short-to-moderate cycles of Type B limit that exposure. Second, the dispersion in Type C was inflated by a small number of low-yielding solid-wood and corn-stalk batches, consistent with the greater handling and loading demands of a sealed inner chamber relative to an open controlled-draft vessel. The two effects compound: a longer, more procedure-dependent cycle offers more chances for an operator-timing deviation to register as an outlier. This reframes the case for each upper-ladder design. The argument for Type C rests on its mean yield, not its consistency, whereas Type B couples an enclosed combustion zone and regulated draft with a forgiving operational envelope, making it the more practical "everyday" choice where consistency, throughput, and capital cost weigh as heavily as peak yield. Because perceived predictability is itself a documented driver of sustained technology use among smallholders [19,56], the repeatability advantage of moderate complexity may matter as much for adoption as the yield advantage of maximal complexity, a consideration rarely surfaced when kilns are ranked on yield alone.

4.5. Comparison with global smallholder systems and the technology ladder

Placed against the global range of small-scale biochar systems, the three kilns span most of the practically accessible performance spectrum. Open burns and earth pits, the default in many smallholder settings, recover only 10%–20% of feedstock mass as biochar [21]; flame-curtain kilns of the kind characterized by Cornelissen et al. [26] typically reach 25%–40%; and retorts occupy the upper band of 30%–50% [32]. Our measured yields of 22%, 32%, and 47% map cleanly onto the lower, middle, and upper segments of this distribution, which both validates the prototypes against independent benchmarks and locates each design within the wider landscape: Type A performs as a well-built flame curtain at the bottom of the curtain-kiln band, Type B in its upper reaches, and Type C at the lower edge of the retort band that has previously required workshop or industrial fabrication. That all three units can be built for under US$1300 places them within the affordability envelope documented for African and Asian smallholder programmes, where [7] reported payback periods of one to three years for drum kilns and rudimentary retorts; under comparable assumptions about biochar value and feedstock supply, the incremental cost of

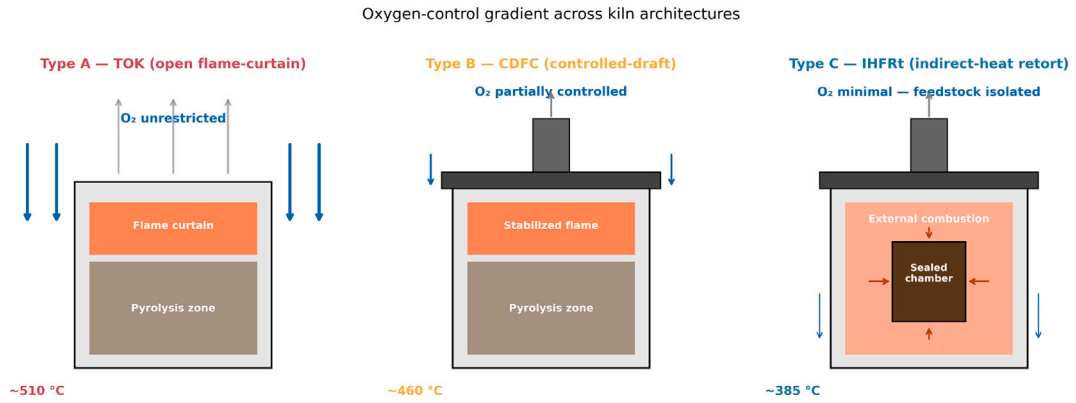


Fig. 9. Conceptual representation of the oxygen-control gradient across kiln architectures. Type A operates with unrestricted oxygen ingress and the highest peak temperatures; Type B introduces a fitted lid and chimney that partially regulate airflow; Type C isolates the feedstock entirely from direct combustion in a sealed inner chamber, enabling oxygen-limited pyrolysis at the lowest peak temperatures.

moving up a rung is recoverable within a modest number of additional cycles, with the case for the upper rungs resting on consistency and product uniformity as much as on capital efficiency. Extension programmes and practice guidelines have made versions of these designs increasingly accessible to on-farm producers [24,25], yet without field-validated benchmarks of the kind reported here the qualitative ordering they convey has been difficult for practitioners to translate into specific performance expectations. The underlying basis for this ordering is the progressive increase in oxygen control across kiln architectures (Fig. 9), which alters the thermal environment experienced by the feedstock and provides a process-based explanation for the observed differences in biochar yield and repeatability.

The more consequential contribution, however, is conceptual. Building on this oxygen-control gradient (Fig. 9), the "technology ladder" has circulated in the smallholder-biochar literature largely as a qualitative ordering, an intuition that more controlled kilns perform better without a quantified account of how much better, why, or relative to which competing factors. The present study converts that intuition into a measured framework: it assigns each rung a yield, a repeatability, a cycle time, and a fabrication cost; it shows that the ordering is driven by a single mechanistic axis (oxygen control acting through peak operating temperature) rather than by feedstock or operation; and it demonstrates that the rungs are separated by effect sizes so large that the kilns do not overlap on yield at all (Fig. 7). As noted above, the quantitative variance partitioning and the GAM-based identification of peak operating temperature as the dominant thermal driver represent, to our knowledge, new evidence at farmer scale, and together they extend previous qualitative framings by quantifying both the size and the mechanism of the steps between rungs. The three designs accordingly occupy distinct, non-dominating positions in a performance space defined by yield, throughput, and affordability (Fig. 8): Type A is the rational choice where capital cost and operator simplicity bind hardest, Type B where consistency and routine throughput matter most, and Type C where high biochar yield or product uniformity justify the longer cycles and added fabrication effort. Rendering the ladder quantitative in this way turns a heuristic into a basis for evidence-based selection that extension services, project developers, and producers can apply directly.

4.6. Implications for decentralized pyrolysis and climate mitigation

The practical headline for decentralized deployment is that a hardware decision, specifically the degree of oxygen control built into a kiln, can roughly double gravimetric yield at modest additional cost, independent of feedstock and largely independent of operator skill. This matters for climate-mitigation programmes built on residue valorization because the quantity of biochar produced per unit of feedstock sets the ceiling on the stable carbon that any downstream accounting can credit; doubling solid-product recovery enlarges that pool from the same biomass base, although the fraction ultimately durable must be established through the elemental and stability characterization this study did not perform. Framing the contribution in yield-and-recovery terms rather than asserting a measured carbon outcome keeps the climate argument defensible while preserving its force: the kilns at the upper rungs convert a far larger share of otherwise-burned residue into a storable product, which is the necessary first step in the value chain that Lehmann et al. [57] describe for biochar-based mitigation.

Because the dominant performance lever is design rather than tightly controlled operating conditions, these benefits are unusually transferable. A locally fabricable, open-source kiln whose yield advantage survives variation in feedstock and operator is well suited to the decentralized, low-supervision contexts in which smallholder carbon projects must function, and the moderation of feedstock-driven variability in the higher-tier designs further lowers the technical threshold for participation in carbon-credit and ecosystem-service markets [57]. The empirical benchmarks reported here, yields, repeatabilities, cycle times, and costs tied to specific, reproducible architectures, provide concrete inputs for the extension training materials, small-scale carbon-project methodologies, and investment due-diligence frameworks that currently lack field-validated performance data for kilns in this class [24,25]. In that sense the study connects a workshop-level fabrication decision to the larger machinery of climate-smart agriculture and global biochar deployment, supplying the kind of quantified, design-anchored evidence those systems require.

4.7. Limitations and future research

Several boundaries of the present study should guide future work. First, we did not characterize biochar physicochemical properties (pH, surface area, elemental ratios, functional groups), which prior work shows are themselves shaped by temperature, residence time, and oxygen control [22]. In this regard, we emphasize that "yield" throughout this study denotes gravimetric mass recovery only and does not imply superior biochar quality. Linking the kiln-specific operating envelopes documented here to product quality is a natural next step, particularly for agronomic relevance across soil and cropping systems.

Second, although the prototypes performed reliably under controlled conditions, real-world operation by farmers introduces additional variability in feeding rate, quench timing, and moisture management that may erode the repeatability advantages observed here. Field trials under farmer-managed operation, combined with structured training of the kind promoted in extension practice guidelines for flame-cap kilns [25], would more accurately characterize the size and stability of the upper rungs of the technology ladder in deployment.

Table 7
Materials, tooling, and equipment for kiln fabrication and operation. Equivalent locally available models or tools may be substituted.

Category	Component	Specification	Purpose/notes
Structural steels	Mild steel sheet	7 ga, 48 × 48 in	High-wear rings and panels; sections exposed to heat and abrasion.
	Mild steel sheet	12 ga, 48 × 48 in	Kiln/retort bodies, lids, and skirts; balance of weight and durability.
	Metal box stock	14 × 20 in	Inner charge chamber for retort (Type C).
	Rebar/iron rod	10–16 mm diameter	Handles, lifting lugs, and local stiffeners.
Fabrication equipment	Press brake	shop unit	Forming cones, flanges, and bends.
	Plate/shear	1/2 in capacity	Straight cuts to size.
	Plasma cutter	shop unit	Curved cuts, ports, and vents.
	MIG welder	Titanium MIG 140 (or equiv.)	Structural seams and brackets.
	Angle grinder	4.5–5 in	Edge preparation and weld cleanup.
	Hand tools	hammers, clamps, squares	Layout, fixturing, and measurement.
Personal protective equipment (PPE)	Welding helmet	auto-darkening	Eye and face protection during welding and ignition.
	Safety apparel	leather gloves, FR clothing	Heat and burn protection.
	Respiratory/eye	dust masks, goggles	Ash and particulate protection.
Operation & testing	Shovels/rakes	–	Loading, leveling, and ash management.
	Water supply	buckets, hoses	Quenching and fire safety.
	Ignition tools	propane torch, matches	Startup and relight.
	Cutting/splitting	chainsaw, wood splitter	Feedstock preparation.
	Sample pans	foil pans	Char collection, cooling, and weighing.
	Measuring aids	scales, tins	Yield and moisture determination.
	Safety gear	fire extinguisher, first-aid kit	On-site safety.

Third, the economic analysis presented here is deliberately a simple capital-cost framing. A full life-cycle and energy-balance assessment, integrating particulate and greenhouse-gas emissions monitoring, would help position low-cost kilns within broader climate-mitigation strategies and provide a stronger basis for inclusion in carbon-credit methodologies [3,54,58,59].

Finally, while the present four-feedstock design is sufficient to establish that kiln architecture, rather than feedstock, governs yield across the tested range, extending the same crossed design to a broader set of residues and to higher-moisture tropical conditions would test the external validity of this architecture-dominance result.

5. Conclusion

This study evaluated three locally fabricable biochar kiln archetypes: an open flame-curtain kiln (Type A; TOK), a controlled-draft flame-curtain kiln (Type B; CDFC), and an indirect-heat retort (Type C; IHFRt) across 60 batch trials and four representative Midwestern agroforestry residues. Mean gravimetric yields rose from 22.3% in Type A to 31.5% in Type B and 46.5% in Type C, with corresponding mean peak temperatures of 510, 460, and 385 °C. Linear mixed-effects modeling and variance partitioning identified kiln architecture as the dominant driver of yield (92%–95% of variance), with feedstock identity contributing less than 1% and the kiln × feedstock interaction being negligible. A generalized additive model showed that peak operating temperature alone explained 92% of the deviance in yield, with a slope of approximately −0.19 percentage points per °C. The yield–time correlation across kilns is weakly positive, reflecting an architecture-level trade-off between throughput and biochar recovery rather than a within-kiln lever. The controlled-draft kiln was the most repeatable design (CV 6.9%), followed by the indirect-heat retort (CV 7.6%) and the open flame-curtain kiln (CV 13.2%); these figures describe gravimetric recovery rather than char quality.

Together these results support a quantitative "technology ladder" for farmer-scale biochar production: Type A for low-cost demonstration and rapid throughput, Type B for reliable everyday production with good consistency, and Type C where high biochar yield and product uniformity justify longer cycles and higher fabrication cost. All three designs can be built in a basic workshop for under US$1300, providing practical fabrication-cost benchmarks for decentralized producers and extension programs. The validated benchmarks reported here provide an empirical basis for evidence-based technology selection, extension training, and the wider deployment of decentralized, smallholder biochar systems in support of residue valorization and climate-mitigation goals.

CRediT authorship contribution statement

Francis Asare: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rado Gazo:** Writing – review & editing, Supervision, Resources, Methodology, Conceptualization. **Francis Wilson Owusu:** Writing – review & editing, Validation, Methodology. **Shristi Shefali Saraugi:** Writing – review & editing, Investigation, Formal analysis. **Winny Routray:** Writing – review & editing, Validation, Methodology. **Jue Mo:** Writing – review & editing, Visualization, Formal analysis. **Shams Rahman Rahmani:** Writing – review & editing, Resources, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.bmf.2026.100059.

Data availability

Data will be made available on request.

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